

Package #160

Impose a controlled biaxial strain field on the borosilicate glass liner during high-temperature annealing (e.g., via differential cooling or external mechanical loading) to refine grain size and strengthen grain boundaries, thereby suppressing grain-boundary sliding

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'grain-boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why do grain-boundary sliding failures limit the service temperature of borosilicate glass reactor liners, and how can the grain structure be refined to fix this?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2025). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Integrated advanced strain engineering combined with thermal annealing to modify grain boundary characteristics

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.